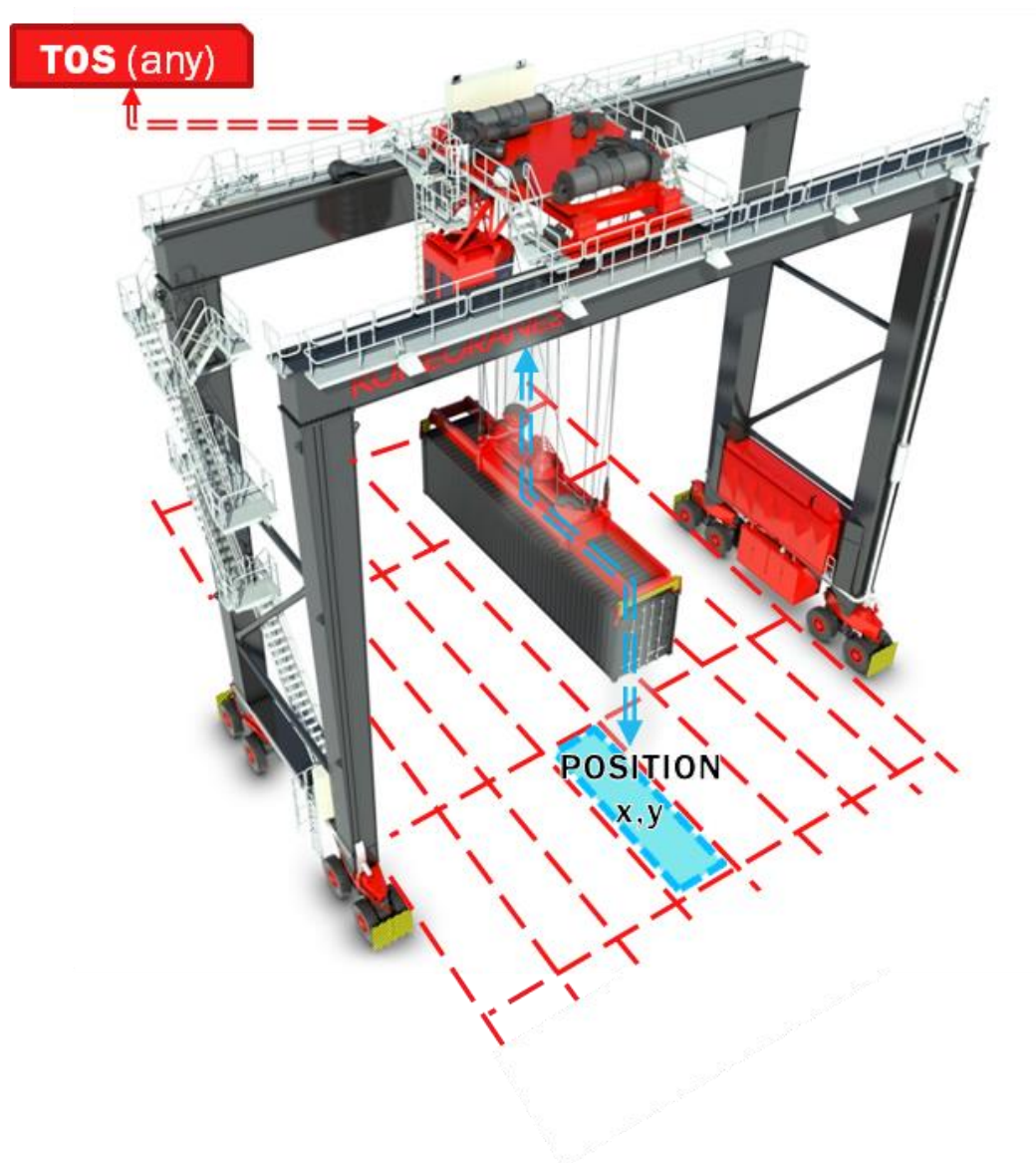


# INTERFACE TO THE TOS

## Technical Description

Yard Process Automation  
21.05.2015  
4.0



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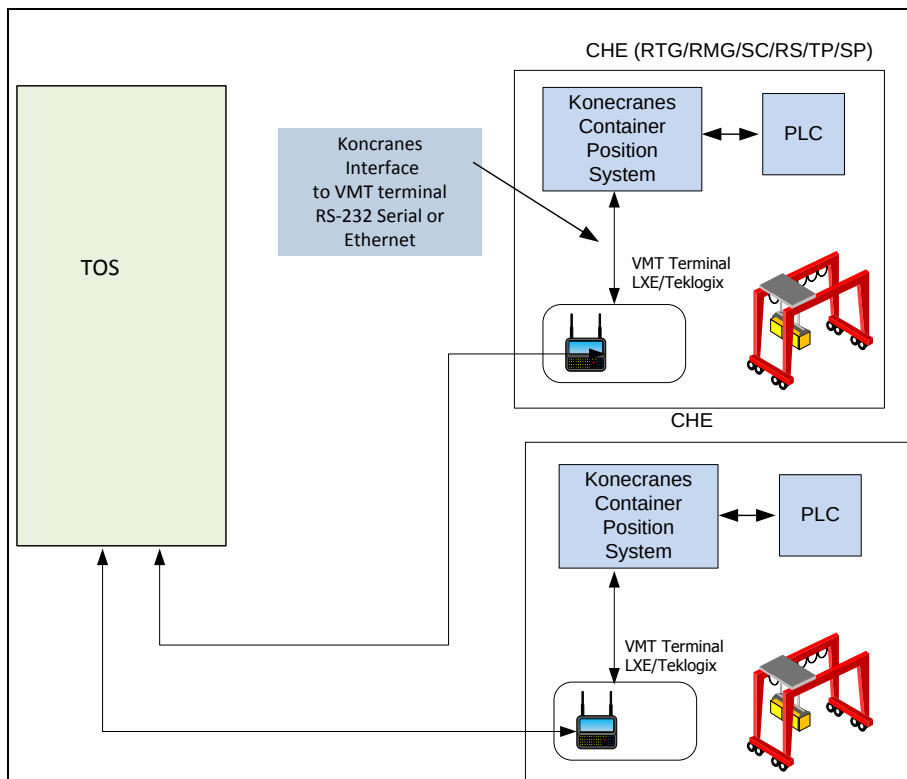
## 1. TERMINOLOGY

|          |                                      |
|----------|--------------------------------------|
| Auto-TOS | Konecranes Auto-TOS reporting system |
| Auto-pos | Konecranes Auto-positioning system   |
| CHE      | Container Handling Equipment         |
| CRC      | Cyclic Redundancy Counter            |
| TOS      | Terminal Operating System            |
| VMT      | Vehicle Mount Terminal               |

## 2. GENERAL DESCRIPTION

The interface to the TOS provides real-time messaging between the Konecranes Auto-TOS reporting system and the TOS. The messages are a mechanism for updating the TOS with real-time container movement information. The messaging is bi-directional: from the Auto-TOS to the TOS, and from the TOS to the Auto-TOS.

The interface to the TOS is implemented using an RS-232 serial connection port or Ethernet port in the Auto-TOS. The connection to the TOS terminal is a direct connection. Alternatively, a fiber optical cable can be used between the Konecranes Auto-TOS and TOS terminals, depending on the distance between the terminals. When Ethernet communication is used, the VMT terminal is connected to crane network



### 3. ESTABLISHING THE CONNECTION

A connection can be established without a protocol. The Auto-TOS sends pick and place messages automatically after power-on. The Auto-TOS sends a heartbeat message once per second (the time period is configurable), so that the TOS can detect any communication failures. The heartbeat message can also be requested from the TOS, and automatic heartbeat sending can be disabled.

### 4. MESSAGE ENCAPSULATION

The following protocol is used to encapsulate messages within the byte-stream in order to e.g. detect the beginning and end of a message. The protocol uses two length bytes and a termination byte. Thus the format of the transmitted packet is:

FFXXNNaaaaaaaaaaaa...aaZZ

Where:

- "FF" indicates "start of data" two byte value hex 0xFFFF (internal use)
- "XX" denotes message length by two byte asciihex, max 255 bytes
- "NN" is a message ID code (Msg-ID)
- "NNaaaaaaaaaaaa...aa" is the message as defined later
- "ZZZZ" is the CRC checksum. All of the bytes in the message, including the "XX", will be added up into a binary word that is two bytes wide, two's complemented. ( $\sim\text{sum}+1$ ). The result is transferred to Asciihex format.

**Note 1:** The message length "XX" is defined as the length of the "NNaaaaaaaaaaaa...aa" part.

**Note 2:** Ascii fields are left-justified and padded with blanks unless otherwise noted. Asciihex fields used for 8-bit and 16-bit Integers are in big-endian format (Message Length, Counter, CRC, etc.).

### 5. AUTO-TOS TO TOS MESSAGES

|    |                              |        |
|----|------------------------------|--------|
| A1 | Heartbeat                    | 0x4131 |
| A2 | Pick/Place Message           | 0x4132 |
| A3 | TOS Message Acknowledgement  | 0x4133 |
| A4 | Job Twistlock Action Request | 0x4134 |

### 6. TOS TO AUTO-TOS MESSAGES

|    |                                  |        |
|----|----------------------------------|--------|
| B1 | Heartbeat Request                | 0x4231 |
| B2 | Auto-TOS Message Acknowledgement | 0x4232 |
| B3 | Job Data Message                 | 0x4233 |
| B4 | Job Cancel Message               | 0x4234 |
| B5 | Job Continue Message             | 0x4235 |

## 7. MESSAGES FROM AUTO-TOS TO TOS FORMAT DESCRIPTIONS

### 7.1. Heartbeat Message (A1) format description

| Offset | #bytes | Type  | Name             | Description                                                                                                                   |
|--------|--------|-------|------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 000    | 2      | Ascii | Msg-ID           | Two byte msg ID ("A1"=0x4131 = 16689)                                                                                         |
| 002    | 6      | Ascii | Version          | Version number                                                                                                                |
| 008    | 6      | Ascii | Time             | Read time (HHMMSS)<br>E.g.: 1:31 PM is represented as 133100                                                                  |
| 014    | 4      | Ascii | Status           | Auto-steering status (OK = "0000", any error: <> 0 (d))                                                                       |
| 018    | 8      | Ascii | Position /       | Block name /"ERROR " /"NO BLOCK" bytes 018 – 025                                                                              |
| 026    | 3      |       |                  | Bay number bytes 026 – 028                                                                                                    |
| 029    | 3      |       |                  | Row number bytes 029 – 031                                                                                                    |
| 032    | 4      |       |                  | Height, Logical value 1...7 bytes 032 – 035                                                                                   |
| 036    | 2      | Ascii | Crane status     | "01" = Crane off<br>"02" = Crane on /idle<br>"03" = Crane running<br>"04" = Crane error / Fatal error                         |
| 038    | 2      | Ascii | GPS status       | "01" = GPS O.K<br>"02" = GPS internal module fault<br>"03" = No satellites<br>"04" = No communication to base station<br>.... |
| 040    | 5      | Ascii | CHE              | CHE id                                                                                                                        |
| 045    | 6      | Ascii | System status    | Reserved for internal use<br>System status information                                                                        |
| 051    | 2      | Ascii | PLC comm. status | "01" = Communication OK<br>"02" = Communication Failure                                                                       |
| 053    | 9      | Ascii | PosX             | Che position x-coordinate                                                                                                     |
| 062    | 9      | Ascii | PosY             | Che position y-coordinate                                                                                                     |
| 071    | 2      | Ascii | Len              | Spreader size 20/30/40/45/T (T=Twin-lift)                                                                                     |
| 073    | 1      | Ascii | Twist lock       | "O" twist lock open<br>"C" twist lock close                                                                                   |

**Note 3:** Active GPS status values depends on the CHE type and hardware configuration

**Note 4:** Status and Crane Status used only with auto-steering RTG.

**Note 5** Status, Cranes status, GPS status, System Status and PLC comm. Status are used mainly for Konecranes diagnostic purposes.

**Note 6:** Height is specified as logical height values '1' to '7'. Konecranes YardIT ContPos reads the height value from PLC and converts it to a logical value. Left justified space filled.

**Note 7: Position information:**

The positions are represented in meters from base station (origo).

X-coordinate increases down on the map and Y-coordinate increases right on the map.

The map is rotated "straight" before calculating these coordinates.

18 bytes reserved for X/Y-position.

9 bytes reserved for both X and Y position.

9 bytes are:

(-)DDDDD.DD (two decimal precision i.e one centimeter which is enough)

The positions can vary from -99999.99 m - 99999.99 meters

examples

-1.24

23.45

127.45

-1256.23

-15674.45

etc.

The strings are padded with spaces in front of the string

## 7.2. Pick/Place Message (A2) format description

| offset | #bytes | Type     | Name             | Description                                                                                       |
|--------|--------|----------|------------------|---------------------------------------------------------------------------------------------------|
| 000    | 2      | Ascii    | Msg-ID           | Two byte msg ID ("A2" = 0x4132=16690)                                                             |
| 002    | 6      | Ascii    | Version          | Version number                                                                                    |
| 008    | 2      | AsciiHex | Counter          | 0-255                                                                                             |
| 010    | 2      | Ascii    | Move type        | 03" = CHE lifts container from ground<br>04" = CHE grounds container                              |
| 012    | 14     | Ascii    | Time             | Read time (YYYYMMDDHHMMSS) E.g.: 1:31 PM<br>on 3 of July 2004 is represented as<br>20040703133100 |
| 026    | 5      | Ascii    | Status           | Auto-steering status (OK = "0000", any error: <> 0<br>(d))                                        |
| 031    | 5      | Ascii    | CHE              | CHE id                                                                                            |
| 036    | 5      | Ascii    | ContWeight       | Container weight in kilograms. Valid in Place mes-<br>sage only                                   |
| 041    | 2      | Ascii    | PosOnTruck       | Optional. Measured position when picking/placing<br>from truck. Measured in spreader up position  |
| 043    | 5      | Ascii    | spare            | [spaces ]                                                                                         |
| 048    | 12     | Ascii    | ContID1          | Optional                                                                                          |
| 060    | 12     | Ascii    | ContID2          | Optional                                                                                          |
| 072    | 1      | Ascii    | Mount            | Always "Y"                                                                                        |
| 073    | 1      | Ascii    | Tag              | Always "N"                                                                                        |
| 074    | 8      | Ascii    | Position1        | Block name /"ERROR " /"NO BLOCK"                                                                  |
| 082    | 3      |          |                  | Bay number                                                                                        |
| 085    | 3      |          |                  | Row number                                                                                        |
| 088    | 4      |          |                  | Height                                                                                            |
| 092    | 8      | Ascii    | Position2        | Block name /"ERROR " /"NO BLOCK"                                                                  |
| 100    | 3      |          | (Filled only for | Bay number                                                                                        |
| 103    | 3      |          | Twin-lift)       | Row number                                                                                        |
| 106    | 4      |          |                  | Height                                                                                            |
| 110    | 2      | Ascii    | Len              | Container size 20/30/40/45/T (T=Twin-lift)                                                        |
| 112    | 6      | Ascii    | System status    | Reserved for internal use<br>System status information                                            |
| 118    | 1      | Ascii    | Pos y/n          | y= use pos coordinates<br>n= use Position1/2 data<br>used in rail operation                       |
| 119    | 9      | Ascii    | PosX             | Position x coordinate                                                                             |
| 128    | 9      | Ascii    | PosY             | Position y coordinate                                                                             |
| 137    | 5      | Ascii    | PosZ             | Position z coordinate                                                                             |
| 142    | 30     | Ascii    | spare            | spare, default space                                                                              |

This message is sent automatically by the Auto-TOS whenever the Container Handling Equipment picks a container up or puts one down. The message format has some optional/extra fields for future expansion, unused at the moment...

For twin-lift operation, the container length field (Len) will be "T", and the Position1 and Position2 fields are both filled. If only one container is handled, the container length field (Len) will be 20 or 40 (feet) and only the Position1-field is filled. The Konecranes Auto-TOS does not use container sizes 30 and 45 at the moment.

Konecranes ContPos resends the Pick/Place message if it does not receive B2 acknowledgement message with time limit. Resending can be disabled in the ContPos configuration.

For Rail car operation the coordinates for rail car and truck can be used in the message. TOS can identify the rail car based on coordinates. In pick and place operation handling the truck the position field is as default "NO BLOCK" and position coordinates are empty.

### 7.3. TOS Message Acknowledgement (A3) format description

| offset | #bytes | Type     | Name    | Description                         |
|--------|--------|----------|---------|-------------------------------------|
| 000    | 2      | Ascii    | Msg-ID  | Two byte msg ID ("A3"=0x4133=16691) |
| 002    | 2      | AsciiHex | Counter | 0-255                               |

Every TOS message B3, B4 and B5 will receive an A3-acknowledgement message. The counter equals the received message counter and identifies the message to be acknowledged.

If the TOS does not receive an acknowledgement within 5 seconds, it should resend the message again.

After 3 retries without a response, a "NO communication with Konecranes system" alarm shall be reported by the TOS.

**Counter:** The TOS will incrementally raise the counter value before every new message, but not for retries. The Konecranes Auto-TOS reporting system will check if the message's "Counter" field was incrementally changed and will deal with it as a new message only when it has changed.

### 7.4. Job Twist lock Action request (A4) format description

| offset | #bytes | Type     | Name      | Description                                                                                 |
|--------|--------|----------|-----------|---------------------------------------------------------------------------------------------|
| 000    | 2      | Ascii    | Msg-ID    | Two byte msg ID ("A4" = 0x4134=16692)                                                       |
| 002    | 6      | Ascii    | Version   | Version number                                                                              |
| 008    | 2      | AsciiHex | Counter   | 0-255                                                                                       |
| 010    | 2      | Ascii    | Move type | "03" = CHE lifts request from ground<br>"04" = CHE container ground request                 |
| 012    | 14     | Ascii    | Time      | Read time (YYYYMMDDHHMMSS) E.g.: 1:31 PM on 3 of July 2004 is represented as 20040703133100 |
| 026    | 5      | Ascii    | Status    | Auto-steering status (OK = "0000", any error: <> 0 (d))                                     |
| 031    | 5      | Ascii    | CHE       | CHE id                                                                                      |
| 036    | 12     | Ascii    | Chassis   | [spaces]                                                                                    |



|     |    |       |                  |                                                                             |
|-----|----|-------|------------------|-----------------------------------------------------------------------------|
| 048 | 12 | Ascii | ContID1          | Optional                                                                    |
| 060 | 12 | Ascii | ContID2          | Optional                                                                    |
| 072 | 1  | Ascii | Mount            | Always "Y"                                                                  |
| 073 | 1  | Ascii | Tag              | Always "N"                                                                  |
| 074 | 8  | Ascii | Position1        | Block name /"ERROR " /"NO BLOCK"                                            |
| 082 | 3  |       |                  | Bay number                                                                  |
| 085 | 3  |       |                  | Row number                                                                  |
| 088 | 4  |       |                  | Height                                                                      |
| 092 | 8  | Ascii | Position2        | Block name /"ERROR " /"NO BLOCK"                                            |
| 100 | 3  |       | (Filled only for | Bay number                                                                  |
| 103 | 3  |       | Twin-lift)       | Row number                                                                  |
| 106 | 4  |       |                  | Height                                                                      |
| 110 | 2  | Ascii | Len              | Container size 20/30/40/45/T (T=Twin-lift)                                  |
| 112 | 6  | Ascii | System status    | Reserved for internal use<br>System status information                      |
| 118 | 1  | Ascii | Pos y/n          | y= use pos coordinates<br>n= use Position1/2 data<br>used in rail operation |
| 119 | 9  | Ascii | PosX             | Position x coordinate                                                       |
| 128 | 9  | Ascii | PosY             | Position y coordinate                                                       |
| 137 | 5  | Ascii | PosZ             | Position z coordinate                                                       |
| 142 | 30 | Ascii | spare            | spare, default space                                                        |

This message is sent automatically by the Auto-TOS every time the Container Handling Equipment tries to pick or place a container with active twist-lock blocking. The message structure is the same as in the pick/place message. Normally the response will be a Job Continue message which unblocks the twistlocks.

## 8.MESSAGES FROM TOS TO AUTO-TOS FORMAT DESCRIPTIONS

### 8.1. Heartbeat request (B1) format description

| offset | #bytes | Type  | Name      | Description                                                                                                   |
|--------|--------|-------|-----------|---------------------------------------------------------------------------------------------------------------|
| 000    | 2      | Ascii | Msg-ID    | Two byte msg ID ("B1" = 0x4231=16945)                                                                         |
| 002    | 1      | Ascii | TWL_block | "0" = Twist locks unblocked<br>"1" = Twist locks unblocked in truck lane<br>"2" = Twist locks blocking active |

The TOS application can control the twistlock blocking/unblocking status when sending the Heartbeat Request. The Auto-TOS sends an A1 message as a response. Normally, the Auto-TOS automatic heartbeat sending function is disabled in this case. Twistlock blocking control can be bypassed momentarily by sending a Job Continue message.

### 8.2. Auto-TOS Message Acknowledgement (B2) format description

| offset | #bytes | Type     | Name    | Description                           |
|--------|--------|----------|---------|---------------------------------------|
| 000    | 2      | Ascii    | Msg-ID  | Two byte msg ID ("B2" = 0x4232=16946) |
| 002    | 2      | AsciiHex | Counter | 0-255                                 |

Every A2 and A4 message will receive a B2 acknowledgement message from the TOS. If the Auto-TOS does not receive an acknowledgement within 2 seconds (time is configurable), it will resend the message.

After 3 retries without a response, a "NO Communication with TOS system" alarm will be reported to the crane's PLC (RTG).

A new A2 message will be sent in response to every new message from the TOS.

When an acknowledgement message is received – the "NO Communication with TOS system" alarm will be turned off and a report will be sent to the crane's PLC (RTG).

**Counter:** The Konecranes Auto-TOS will incrementally raise the counter value before every new message, except with retries. The TOS will check if the message's "Counter" field was incrementally changed and will deal with it as a new message only if it has changed. If the counter stays the same, it means that the Auto-TOS did not receive the acknowledgement message: no new e.g. pick/place action was detected.



### 8.3. Job Data Message (B3) format description

| Offset | #bytes | Type     | Name                                              | Description                                                                                                                                                                                |
|--------|--------|----------|---------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 000    | 2      | Ascii    | Msg-ID                                            | Two byte msg ID ("B3" = 0x4233=16947)                                                                                                                                                      |
| 002    | 6      | Ascii    | Version                                           | Version number                                                                                                                                                                             |
| 008    | 2      | AsciiHex | Counter                                           | 0-255                                                                                                                                                                                      |
| 010    | 14     | Ascii    | Time                                              | Read time (YYYYMMDDHHMMSS) E.g.: 1:31 PM on 3 of July 2004 is represented as 20040703133100                                                                                                |
| 024    | 5      | Ascii    | CHE                                               | CHE id                                                                                                                                                                                     |
| 029    | 12     | Ascii    | ContID1                                           | Container ID                                                                                                                                                                               |
| 041    | 12     | Ascii    | ContID2                                           | Container ID 2 (Filled only for Twin-lift)                                                                                                                                                 |
| 053    | 8      | Ascii    | Lift-Position1                                    | Block name bytes 053 – 060                                                                                                                                                                 |
| 061    | 3      |          |                                                   | Bay number bytes 061 – 063                                                                                                                                                                 |
| 064    | 3      |          |                                                   | Row number bytes 064 – 066                                                                                                                                                                 |
| 067    | 4      |          |                                                   | Height, Logical value 1...7 bytes 067 – 070                                                                                                                                                |
| 071    | 8      | Ascii    | Lift-Position2<br>(Filled only for<br>Twin-lift)  | Block name bytes 071 – 078                                                                                                                                                                 |
| 079    | 3      |          |                                                   | Bay number bytes 079 – 081                                                                                                                                                                 |
| 082    | 3      |          |                                                   | Row number bytes 082 – 084                                                                                                                                                                 |
| 085    | 4      |          |                                                   | Height, Logical value 1...7 bytes 085 – 088                                                                                                                                                |
| 089    | 8      | Ascii    | Place-Position1                                   | Block name bytes 089 – 096                                                                                                                                                                 |
| 097    | 3      |          |                                                   | Bay number bytes 097 – 099                                                                                                                                                                 |
| 100    | 3      |          |                                                   | Row number bytes 100 – 102                                                                                                                                                                 |
| 103    | 4      |          |                                                   | Height, Logical value 1...7 bytes 103 – 106                                                                                                                                                |
| 107    | 8      | Ascii    | Place-Position2<br>(Filled only for<br>Twin-lift) | Block name bytes 107 – 114                                                                                                                                                                 |
| 115    | 3      |          |                                                   | Bay number bytes 115 – 117                                                                                                                                                                 |
| 118    | 3      |          |                                                   | Row number bytes 118 – 120                                                                                                                                                                 |
| 121    | 4      |          |                                                   | Height, Logical value 1...7 bytes 121 – 124                                                                                                                                                |
| 125    | 2      | Ascii    | Len                                               | Container size 20/30/40/45/T (T=Twin-lift)                                                                                                                                                 |
| 127    | 2      | Ascii    | PosOnTruck                                        | Container position on truck or planned position to truck starting from back of the truck<br>1. 20" rear position<br>2. 20" middle position or ( 40/45 container)<br>3. 20 " front position |
| 129    | 1      | Ascii    | Lift PosUsed                                      | "Y" = Pos coordinates, "N" = Logical Lift position, used in rail operation                                                                                                                 |
| 130    | 1      | Ascii    | Place Truck                                       | "Y"=truck, "N"= Rail Car                                                                                                                                                                   |
| 131    | 9      | Ascii    | Lift posX                                         | Lift x coordinate                                                                                                                                                                          |
| 140    | 9      | Ascii    | Lift posY                                         | Lift y coordinate                                                                                                                                                                          |
| 149    | 5      | Ascii    | Lift posZ                                         | Lift z coordinate                                                                                                                                                                          |
| 154    | 1      | Ascii    | Place PosUsed                                     | "Y" = Pos coordinates, "N"= Logical Lift position, used in rail operation                                                                                                                  |
| 155    | 1      | Ascii    | Place Truck                                       | "Y"=truck, "N"= Rail Car                                                                                                                                                                   |
| 156    | 9      | Ascii    | Place posX                                        | Place x coordinate                                                                                                                                                                         |
| 165    | 9      | Ascii    | Place posY                                        | Place y coordinate                                                                                                                                                                         |
| 174    | 5      | Ascii    | Place posZ                                        | Place z coordinate                                                                                                                                                                         |
| 179    | 20     | Ascii    | spare                                             | spare, default space                                                                                                                                                                       |

This message is sent by the TOS every time the Container Handling Equipment (CHE) driver selects a job, and the job is assigned to the CHE. This data is needed for automated CHE operations (e.g. Auto-Stop).

The Lift-Position1, Lift-Position2, Place-Position1 and Place-Position2 fields have the same format as Position1 and Position2 in the Pick/Place message. **The container length field (Len) is important for Auto-positioning: it's needed to calculate the stopping position correctly (also for twin-lift operation).** For twin-lift operation, the container length field (Len) is

“T” and all four Position fields are filled. If only one container is handled, only the Lift-Position1 and Place-Position1 fields are used.

For Rail car operation the coordinates for rail car and truck can be used. Truck position is calculated based on rail car position.

#### 8.4. Job Cancel Message (B4) format description

| Offset | #bytes | Type     | Name    | Description                                    |
|--------|--------|----------|---------|------------------------------------------------|
| 000    | 2      | Ascii    | Msg-ID  | Two byte msg ID (“ <b>B4</b> ” = 0x4234=16948) |
| 002    | 2      | AsciiHex | Counter | 0-255                                          |

The TOS can cancel an ongoing job by sending the Job Cancel Message. Acknowledgement is required. Job can't be cancelled if the twistlocks are engaged and container has been picked. A PICK message has been sent to TOS. .

#### 8.5. Job Continue Message (B5) format description

| Offset | #bytes | Type     | Name    | Description                                    |
|--------|--------|----------|---------|------------------------------------------------|
| 000    | 2      | Ascii    | Msg-ID  | Two byte msg ID (“ <b>B5</b> ” = 0x4235=16949) |
| 002    | 2      | AsciiHex | Counter | 0-255                                          |

The Job Continue Message communicates that the current job state can continue. Acknowledgement is required.

### 9. INDICATION BASED ON HEARTBEAT GPS STATUS

GPS status

“0001” = GPS OK (Green indication, system OK)

“0002” = GPS internal module fault (red indication, Alarm)

“0003” = No satellites (red indication, Alarm)

“0004” = No communication to base station (red indication, Alarm)

## 10. VALUES OF "POSITION" FIELDS

The Auto-TOS must conform to the customer's names for the position. The yard mapping and Auto-TOS configuration specify the valid values for each position field.

### 10.1. Position notation

**Y12     , 250, 001, 1**

- Where the Y12 **Area/Block** code is left-justified and padded with 5 spaces
- The Bay number max 3 characters, left-justified and space-padded
- The Row number max 3 characters left-justified and space-padded
- The height (Tier) is defined as logical value 1 to 7 left-justified and space-padded

### 10.2. "No block" in position field

"NO BLOCK" is announced if the Container Handling Equipment is out of a defined area: Out of block, bay and row.

### 10.3. "Error" in the position field

If "GPS OK" is not displayed as the GPS status, or if there is a height measurement error, the text "ERROR " (8 characters) is displayed in the "Block name" bytes. The remaining position bytes will be filled with spaces. This means that Bay, Row and Height will all be spaces.

## 11. ASCI HEX PRESENTATION

The following integer values shall be sent and transformed to asciihex format before transferred over the interface.

|      |                                |         |
|------|--------------------------------|---------|
| ZZZZ | CRC (16-bit integer)           | 4 bytes |
| XX   | Message Length (8-bit integer) | 2 bytes |
|      | Counter (8-bit integer)        | 2 bytes |

Each byte is divided into 4 bit sections. Each section receives a corresponding ascii value.

| Section hex | Ascii/hex value | Section hex | Ascii/hex value |
|-------------|-----------------|-------------|-----------------|
| 0           | 0 0x30          | 8           | 8 0x38          |
| 1           | 1 0x31          | 9           | 9 0x39          |
| 2           | 2 0x32          | A           | A 0x41          |
| 3           | 3 0x33          | B           | B 0x42          |
| 4           | 4 0x34          | C           | C 0x43          |
| 5           | 5 0x35          | D           | D 0x44          |
| 6           | 6 0x36          | E           | E 0x45          |
| 7           | 7 0x37          | F           | F 0x46          |

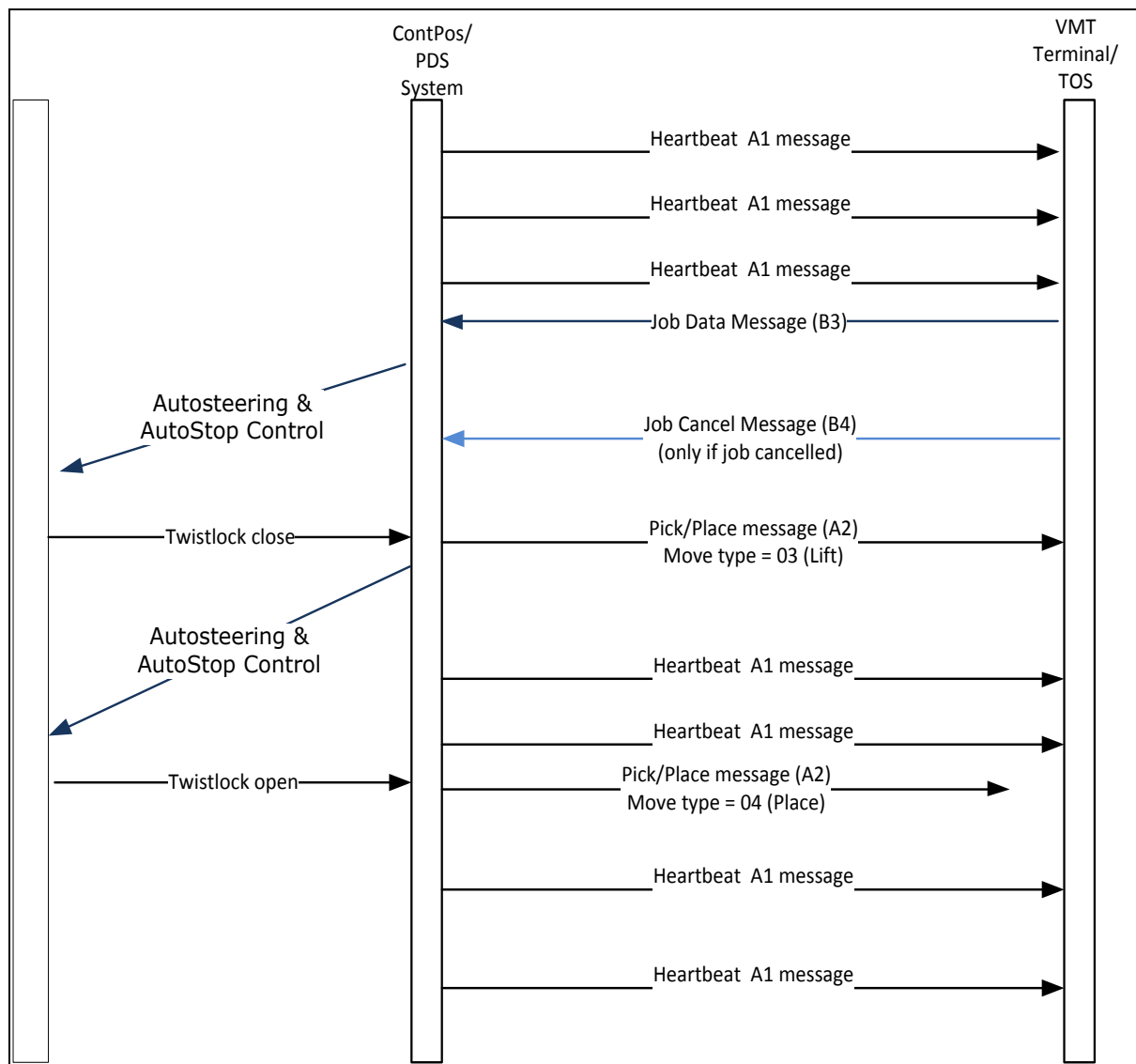
E.g. Counter value 100

The hex value is 0x64 and the asciihex value in the Counter field is 64 (0x36 0x34).

The heartbeat message size is 53 bytes. The value is 35 in hexadecimal format and the value in the message is presented in two bytes as (0x33 0x35).  
The 16-bit integer is in big endian format, and the asciihex is in the same format, MSB section first.

## 12. MESSAGE SEQUENCES EXAMPLES

Message Interfaces as of example in RTG AutoStop/ContPos



Konecranes AutoStop message sequence example

## 13. VERSION HISTORY

| Revision | Date       | Description of change                                                                                                                                                                                                                                                         | Editor |
|----------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|
| 1.00     | 5.9.2007   | First version                                                                                                                                                                                                                                                                 | MJM    |
| 1.01     | 9.7.2007   | Integer fields converted to asciihex to support narrow band connections. Msgld's changed to ascii. Che id added to heartbeat message. System status bytes added at the end of heartbeat and pick/place message.                                                               | MJM    |
| 2.0      | 22.4.2009  | Twist lock control functionality additions. New messages Job Twist lock Action request, Job Cancel Message and Job Continue Message. ACK message changed to include all messages, which requires acknowledgement. New file in message Heartbeat request and Heartbeat message | MJM    |
| 2.0b     | 13.12.2013 | Word template changed to newest Konecranes template. Feature and organization names updated.                                                                                                                                                                                  | TG     |
| 2.1      | 17.04.2014 | Ethernet support included. Language checked                                                                                                                                                                                                                                   | MJM    |
| 3.0      | 25.04.2014 | Heartbeat message field added. CHE position coordinates (PosX, PosY) + spreader and twist lock info                                                                                                                                                                           | MJM    |
| 3.1      | 3.12.2014  | Container position on truck added to job message B3                                                                                                                                                                                                                           | MJM    |
| 3.2      | 15.4.2015  | Container weight added to A2 messages. Support for Gyro system removed.(Top-pick/Side-pick )                                                                                                                                                                                  | MJM    |
| 3.3      | 21.4.2015  | Container position on truck added to A2 message                                                                                                                                                                                                                               | MJM    |
| 4.0      | 21.5.2015  | Rail operation fields added to A2,A4 and B3 messages. Also extra space reserved for future use.                                                                                                                                                                               | MJM    |

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